

Section 9

INTRO TO SE

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State Diagram

is a type of UML diagram used to show how a system or object changes its state based on events.

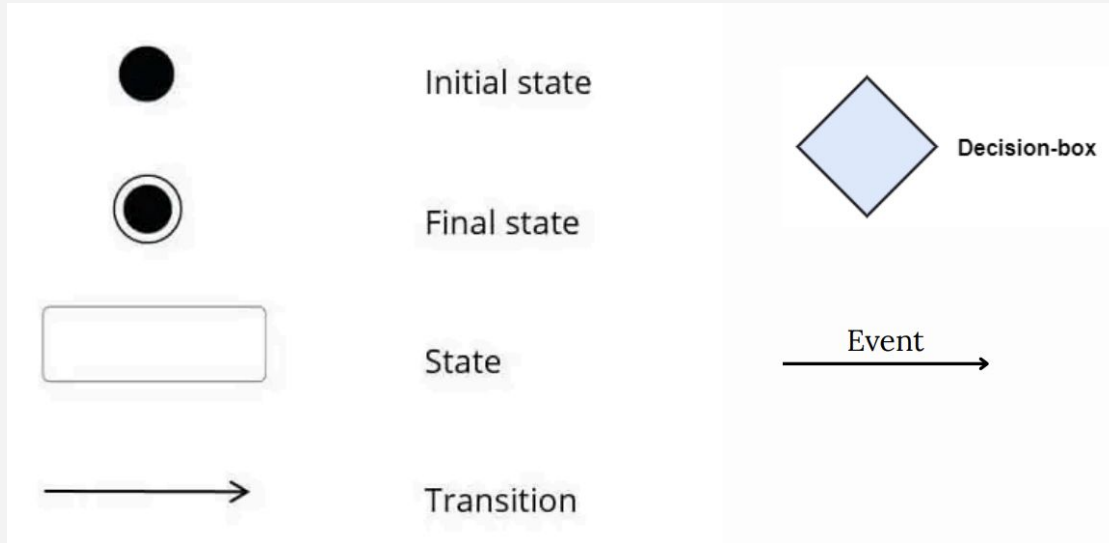
A **state diagram** represents:

- Different **states** an object can be in
- **Transitions** between those states
- **Events** that trigger those transitions

Think of it like:

“What happens next depending on what occurs?”

State Machine Symbols & Notations

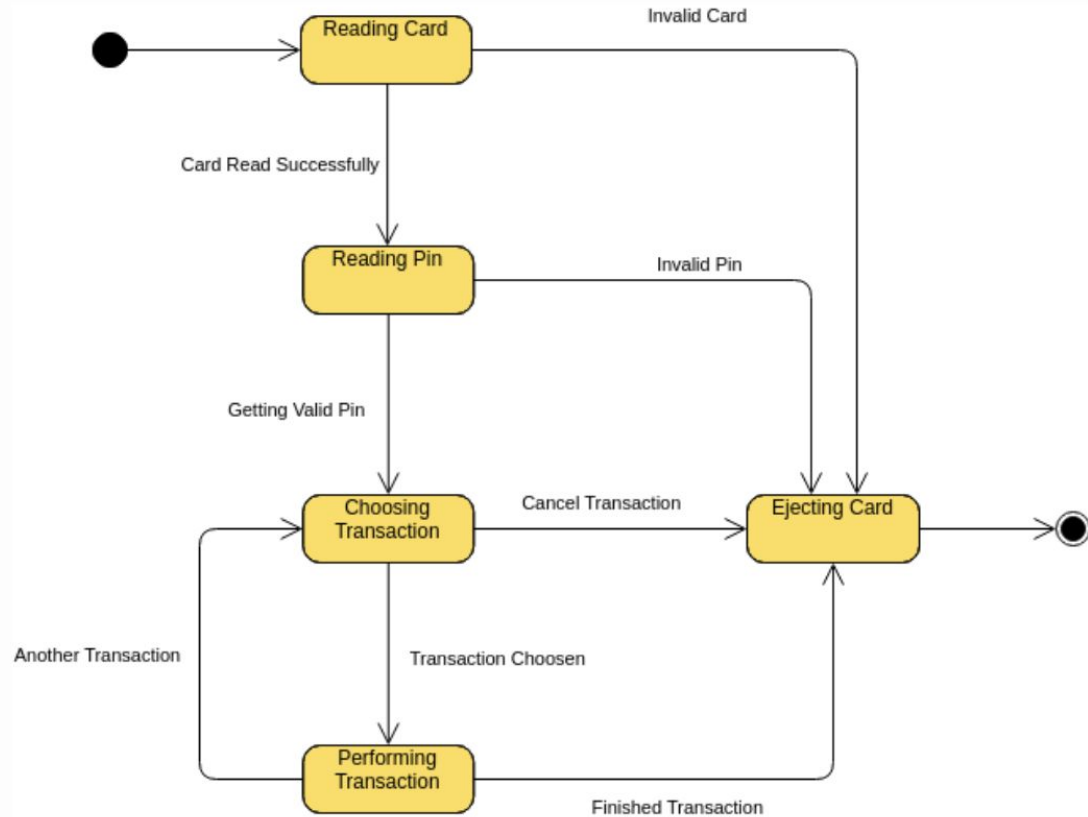


Question:

An **ATM system** allows users to perform banking transactions using a card and PIN.

The system works as follows:

1. When a card is inserted, the system starts by **reading the card**.
 - If the card is **invalid**, the system ejects the card and ends the process.
 - If the card is **valid**, the system proceeds to read the PIN.
2. The system then enters the **PIN reading state**.
 - If the PIN is **invalid**, the card is ejected and the process ends.
 - If the PIN is **valid**, the user can choose a transaction.
3. In the **transaction selection state**, the user can:
 - Choose a transaction → the system performs the transaction
 - Cancel the transaction → the card is ejected
4. During the **transaction execution state**:
 - After completing a transaction, the user may:
 - Perform **another transaction** (return to transaction selection), or
 - Finish → the card is ejected
5. Finally, the system reaches the **end state** after ejecting the card.

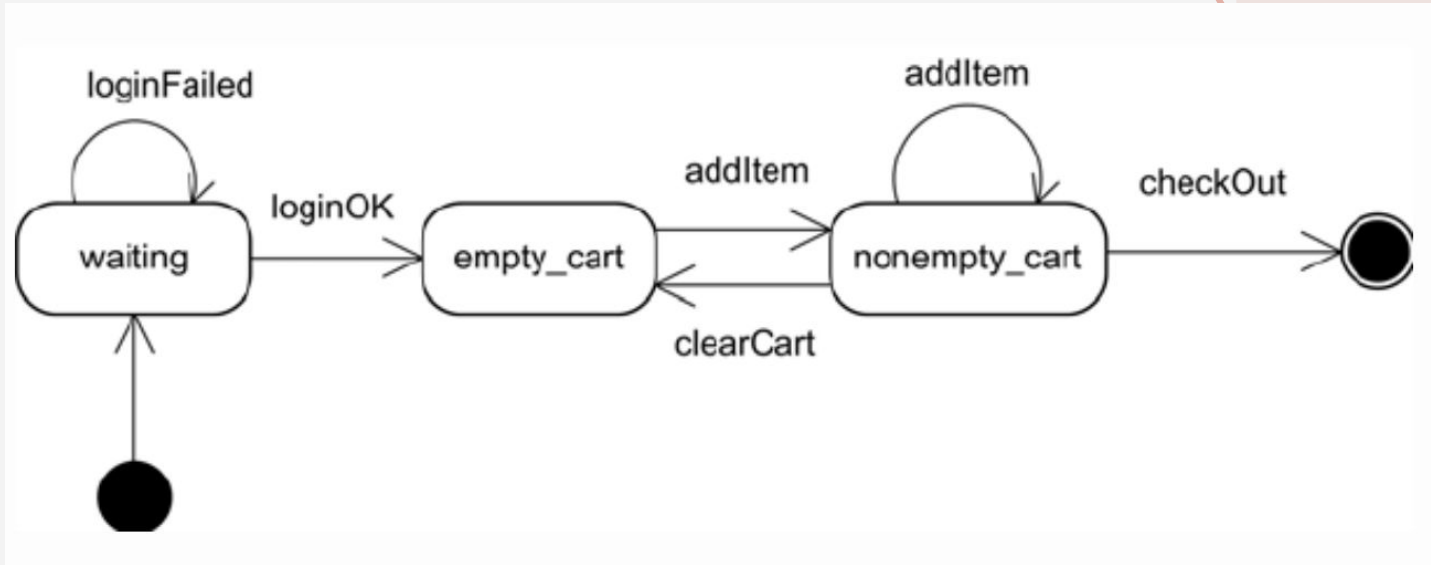


Question:

An **online shopping system** manages a user's login and shopping cart behavior.

The system operates as follows:

1. Initially, the system is in the **waiting state**.
 - If the login **fails**, the system remains in the *waiting* state.
 - If the login is **successful**, the system moves to the *empty cart* state.
2. In the **empty cart state**:
 - When the user **adds an item**, the system transitions to the *non-empty cart* state.
3. In the **non-empty cart state**:
 - The user can continue adding items (the system stays in the same state).
 - The user can **clear the cart**, returning the system to the *empty cart* state.
 - The user can **checkout**, which ends the process.



Sequence Diagram

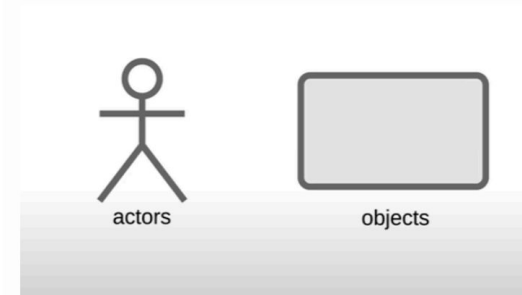
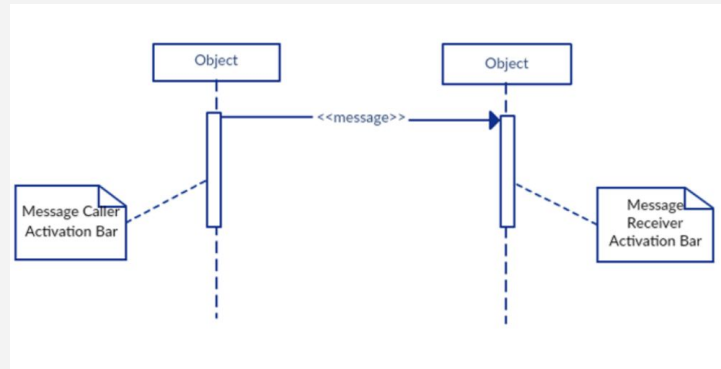
It is type of UML Diagrams that shows how objects and classes in system interact with each other.

This diagram shows interaction in the order they take place.

THEY SHOW THE SEQUENCE OF EVENTS

Sequence Diagram Parts:

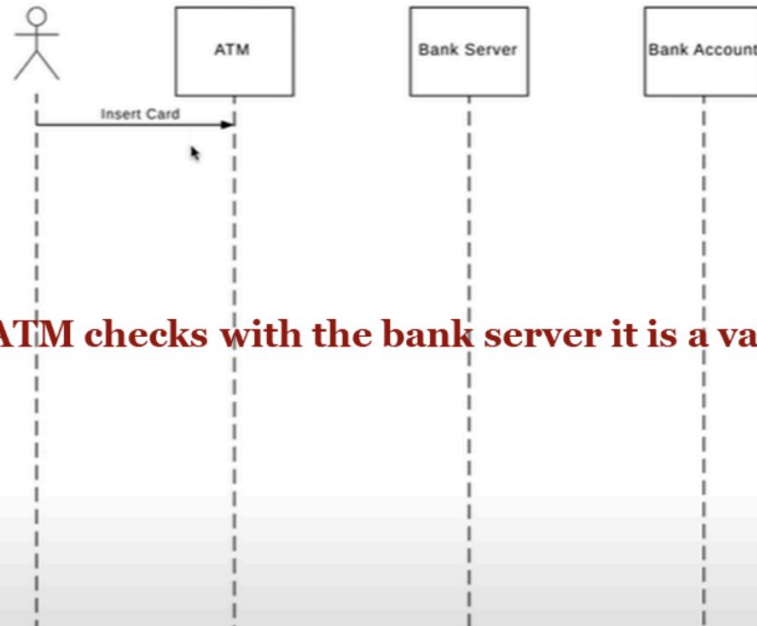
1. Actors
2. Objects
3. Lifeline
4. Messages



Example .01 - ATM Machine

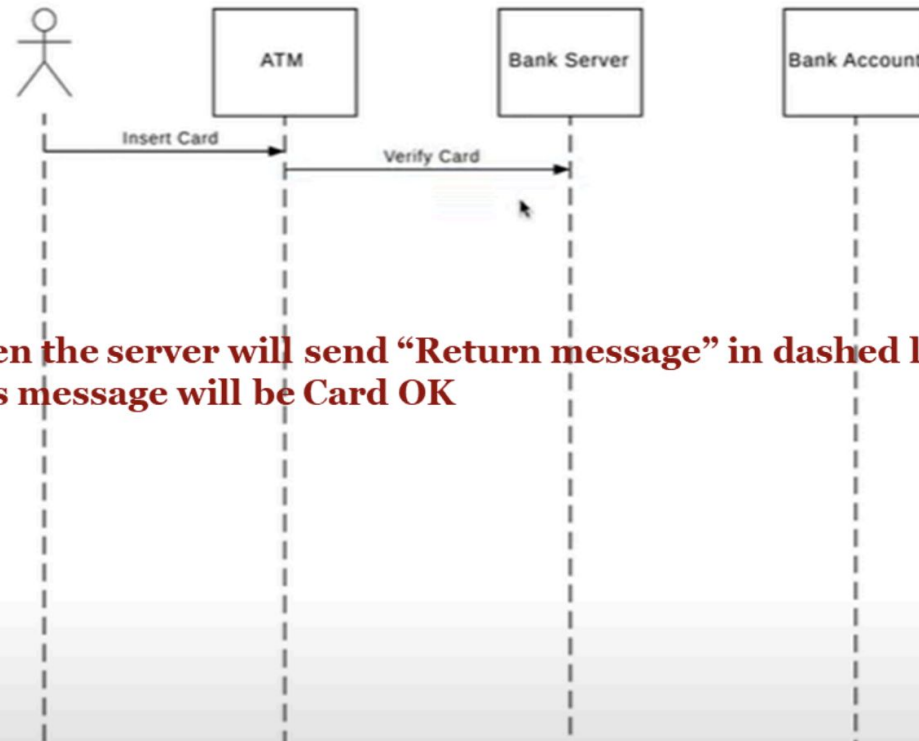
What are the parts involved in ATM Transaction?

1. A person go to an ATM To get Money
2. The ATM communicates to a bank server and that person bank account is then accessed
3. The ATM Bank server and bank account are all part of ATM System



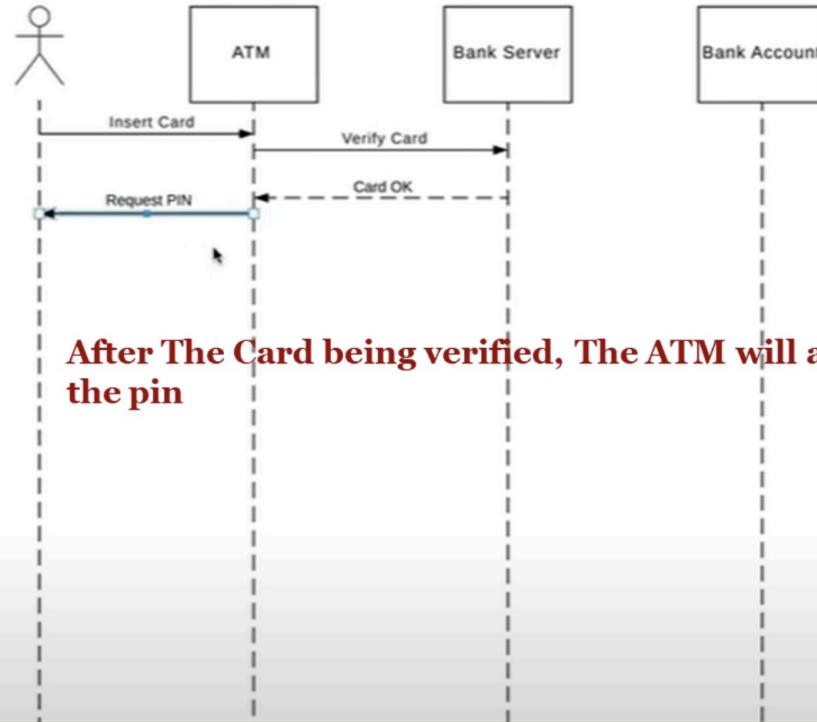
Then ATM checks with the bank server it is a valid Card

Example .01 - ATM Machine



**Then the server will send “Return message” in dashed line
this message will be Card OK**

Example .01 - ATM Machine

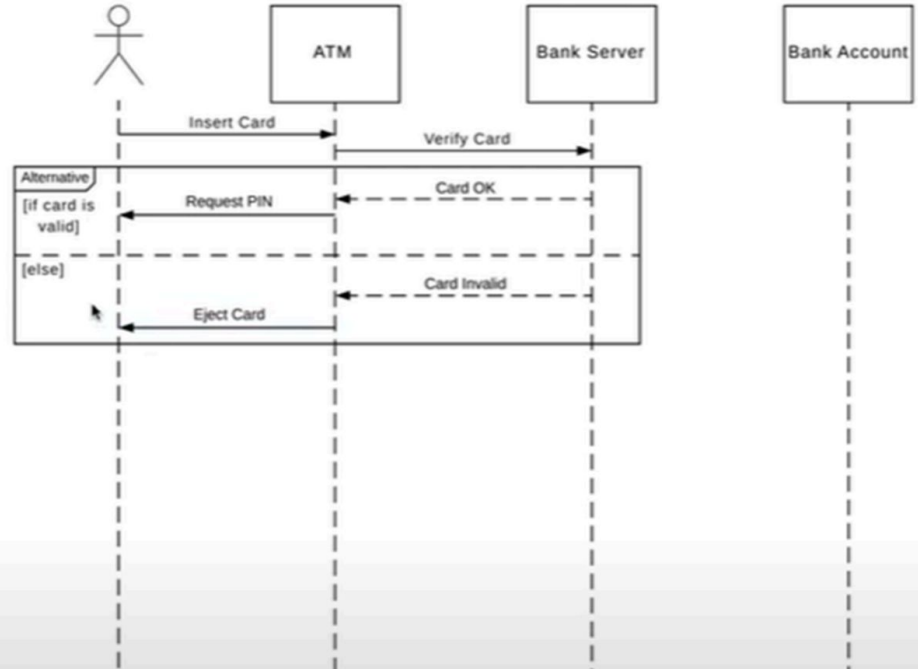


After The Card being verified, The ATM will ask the actor for the pin

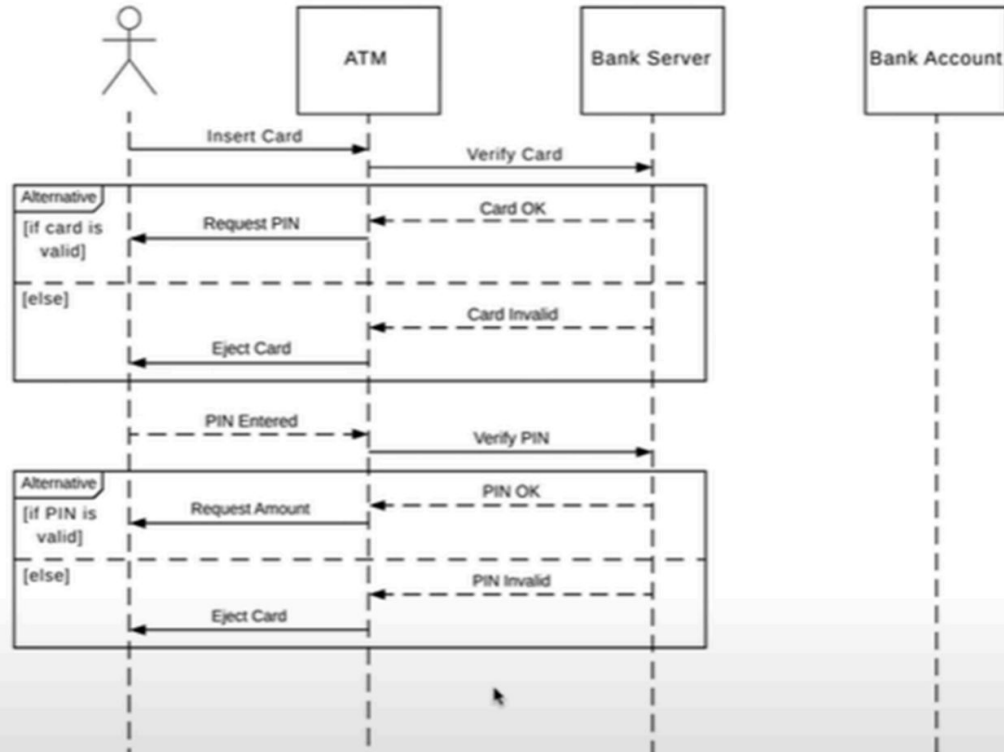
Example .01 - ATM Machine

Alternative Frame

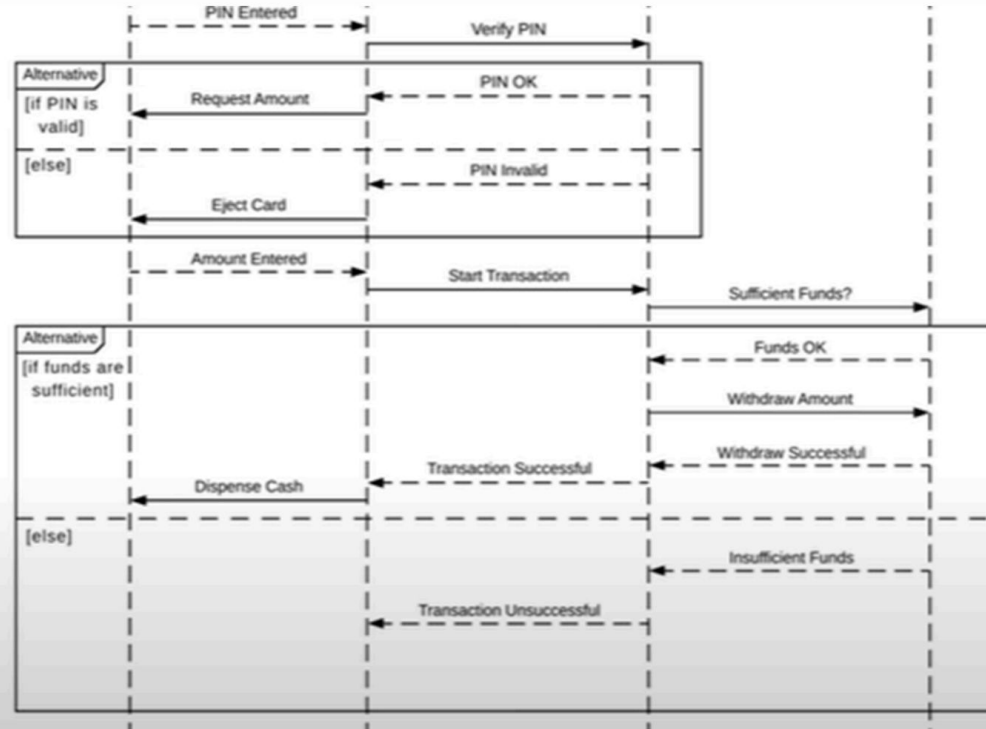
This is used when you have to choose a path between two or more messages sequences



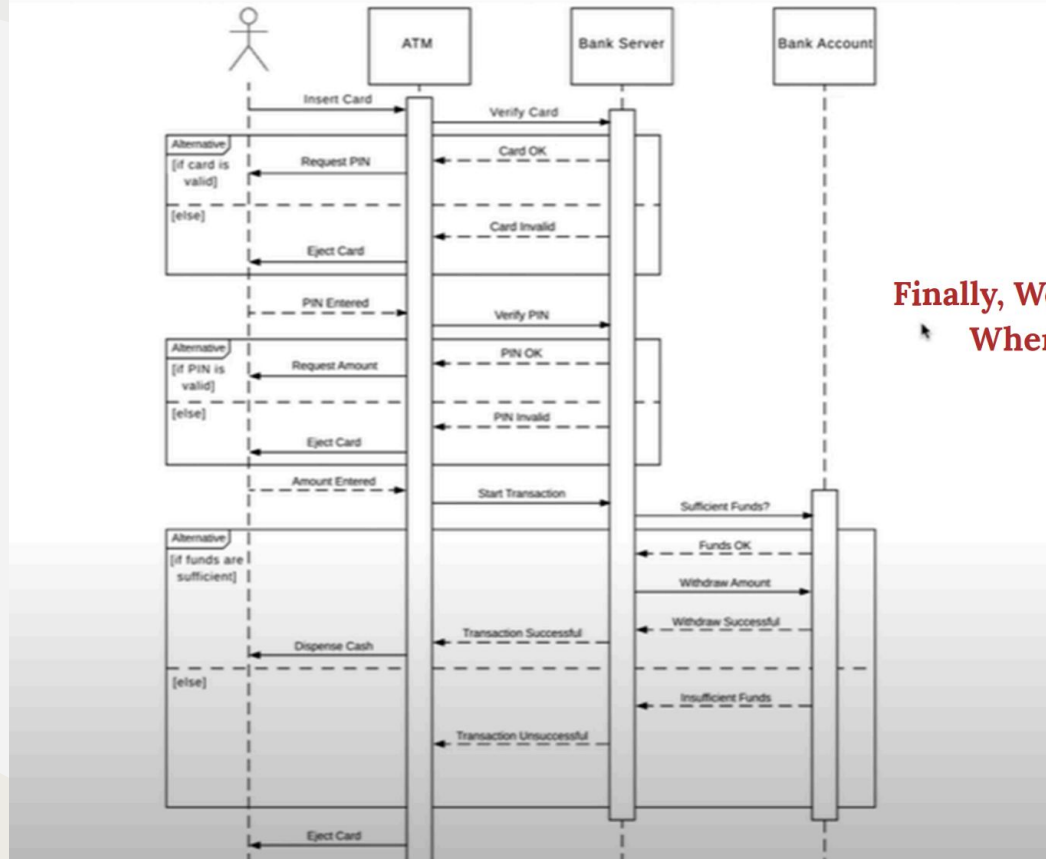
Example .01 - ATM Machine



Example .01 - ATM Machine



Example .01 - ATM Machine



Finally, We add the activation box to show
When and How long an object is
performing a process

